E4.3.1 X-ray attenuation and half-value thickness

A monoenergetic narrow beam in uniform material decreases exponentially in intensity. Half-value thickness halves intensity. Each additional such thickness halves what remains. A larger attenuation coefficient means smaller half-value thickness. Coefficient and thickness units must cancel. The model does not predict a sudden complete stop at finite thickness.

$$I = I_0 e^{-\mu x}$$

$$x_{1/2} = \frac{\ln 2}{\mu}$$

Uniform medium and monoenergetic narrow beam: I_0: incident intensity; I: transmitted intensity; μ: linear attenuation coefficient; x: thickness. μx must be dimensionless (e.g. cm^{-1} with cm).

$x_{1/2}$: half-value thickness; μ: linear attenuation coefficient. Match units, e.g. cm and cm^{-1}; natural logarithm.

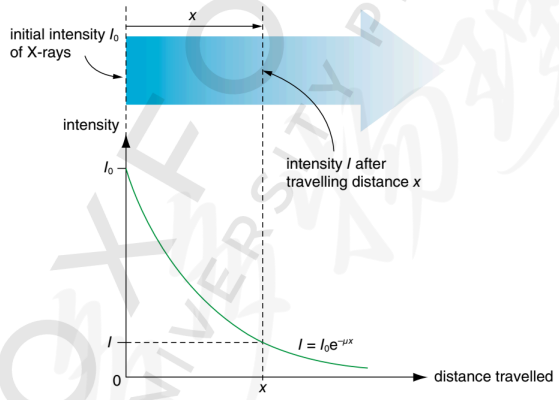


Fig 3.1e Change in intensity over distance travelled.

Intensity decreases exponentially with thickness. Equal additional thicknesses reduce the intensity by equal fractions, not equal absolute amounts.

E4.3.2 Factors affecting X-ray attenuation

Attenuation depends on composition, density and photon energy. With similar composition and energy, greater density generally means stronger attenuation. In common diagnostic-energy ranges away from absorption edges, higher-energy photons usually penetrate more readily. These are conditional trends, not universal rules across all materials and energies.

$$\rho \uparrow \quad \Longrightarrow \quad \mu \uparrow$$

$$E_\gamma \uparrow \Longrightarrow \mu \downarrow \Longrightarrow \text{penetration} \uparrow$$

For otherwise comparable material and photon energy, greater density generally increases linear attenuation. Composition and photon energy also matter.

General trend over diagnostic X-ray ranges away from absorption edges, for the same material. Not a universal monotonic law across every photon energy.

E4.3.3 Attenuation through several layers

Each layer's transmitted intensity becomes the next layer's incident intensity. Multiply transmission fractions, or add each coefficient-times-thickness term in the exponent. Do not add attenuation percentages directly. Measure each thickness along the actual beam path.

$$I = I_0 e^{-\mu_1 x_1} e^{-\mu_2 x_2} \cdots$$

$$I = I_0 e^{-\sum_i \mu_i x_i}$$

Sequential layers: μ_i: layer attenuation coefficient; x_i: thickness along beam. Each μ_i x_i is dimensionless; exponential narrow-beam model.

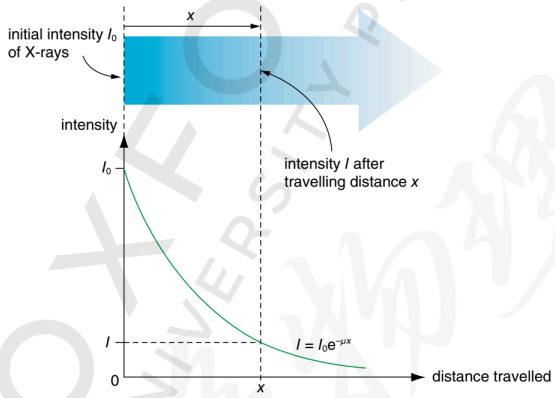


Fig 3.1e *Change in intensity over distance travelled.*

Intensity decreases exponentially with thickness. Equal additional thicknesses reduce the intensity by equal fractions, not equal absolute amounts.

E4.3.4 Brightness and darkness in X-ray images

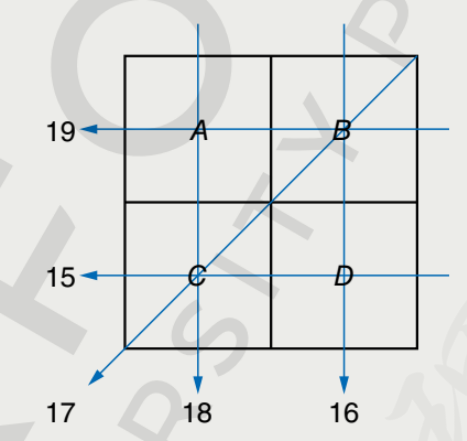
Tissues attenuate X-rays differently, creating contrast in detected transmission. In the conventional textbook display, strongly attenuating structures such as bone appear light and less attenuating regions dark. Digital displays can be inverted, so check the scale. Appearance depends on total path attenuation, not one tissue's density alone.

E4.3.5 Contrast agents

An organ is difficult to distinguish when its attenuation resembles surrounding tissue. An appropriate contrast agent changes attenuation in the chosen region, increasing transmitted-intensity differences and contrast. This improves tissue differentiation, not pixel count or the instrument's intrinsic spatial resolution.

E4.3.6 CT reconstruction from multiple angles

Computed tomography measures attenuation projections through a section from many directions and reconstructs the distribution of attenuation coefficients computationally. Multiple views separate tissues that overlap in an ordinary projection.



Each beam records attenuation summed along its path. Measurements at several angles constrain the individual regions A–D used to reconstruct the slice.

E4.3.7 Pixel area and matrix size

An image matrix states horizontal and vertical pixel counts. For square pixels of area a, side length is $\sqrt{a}$. Express image width W, height H and pixel size in matching units. The matrix is $\frac{W}{\sqrt{a}} \times \frac{H}{\sqrt{a}}$. Multiply the counts for total pixels. A 1024 × 1024 matrix contains 1,048,576 pixels; 1024 is not a physical side length.

E4.3.8 Radionuclide imaging

A radioactive tracer distributes according to physiological processes within the body or target organ. Detectors measure gamma rays emitted from inside to reconstruct tracer distribution and reveal function. Ordinary X-ray imaging instead uses an external source and measures transmission through the body.

A tracer should reach or be taken up by the target region and emit radiation that can escape the body for detection. Its half-life should allow the examination while limiting irradiation afterwards. Activity and amount must provide a useful signal with an appropriate radiation dose.

E4.3.9 Physical half-life

Physical half-life is the time for a nuclide's undecayed population to halve through nuclear decay alone. It is a property of the nuclide and excludes biological removal. Tracer activity in the body can decrease faster because decay and biological clearance operate together.

E4.3.10 Biological half-life

Biological half-life is the time for metabolism, excretion or other biological processes to remove half a substance from the body or target region, independently of nuclear decay. It depends on the substance and physiology. Combining it with physical half-life requires approximately independent exponential decreases.

E4.3.11 Effective half-life

A tracer decreases in the body through both radioactive decay and biological removal. Physical half-life t_{p} describes decay; biological half-life t_{b} describes removal by metabolism, excretion and related processes. Effective half-life t_{e} describes their combined result.

If both decreases are approximately independent and exponential, their removal rates add, giving the reciprocal-sum relation below. Acting together makes the effective half-life shorter than either finite half-life alone. Use the same time units throughout.

$$\frac{1}{t_e} = \frac{1}{t_p} + \frac{1}{t_b}$$

$$t_e = \frac{t_p t_b}{t_p + t_b}$$

Effective half-life t_{e}; physical half-life t_{p}; biological half-life t_{b}. Independent exponential processes; all three times in identical units. t_{e} is less than either finite component half-life.

E4.3.12 Hot and cold spots in radionuclide images

Hot spots show relatively high tracer uptake and cold spots relatively low uptake. Interpretation depends on the tracer and normal distribution. Textbook greyscale makes hot spots darker and cold spots lighter, but other scales differ. The names do not indicate temperature, nor does every hot spot imply the same diagnosis.

E4.3.13 Equivalent dose and radiation weighting

Absorbed dose D is energy absorbed per kilogram, measured in Gy. Equal absorbed doses of different radiation types can have different biological effects. Equivalent dose sums doses multiplied by radiation weighting factors: $H = \sum wD$, in Sv. The textbook uses w = 20 for alpha and w = 1 for beta and gamma. Effective dose additionally accounts for tissue weighting.